1. Using Auxiliary Data & Ratio Estimation (SRS)

Notation (SRS & ratio)

N pop. size; n SRS size; Y_i, X_i values for unit i;
$\tau_y = \sum_{i=1}^N Y_i$; $\tau_x = \sum_{i=1}^N X_i$; $\mu_y = \tau_y/N$; $\mu_x = \tau_x/N$; $\bar{Y} = \frac{1}{n}\sum Y_i$;
$\bar{X} = \frac{1}{n}\sum X_i$; $R = \tau_y/\tau_x = \mu_y/\mu_x$; $\hat{R} = \bar{Y}/\bar{X}$;
$\sigma_y^2, \sigma_x^2, \sigma_{xy}$ population var/cov; $\hat{\sigma}_y^2 = \frac{1}{n-1}\sum(Y_i - \bar{Y})^2$; $\hat{\sigma}_x^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$;
$Z_i = Y_i - RX_i$; $\sigma_r^2 = \frac{1}{N}\sum(Y_i - RX_i)^2$; $\hat{cv}_x = \hat{\sigma}_x/\bar{X}$; $\hat{cv}_y = \hat{\sigma}_y/\bar{Y}$;
Auxiliary X is easy/cheap and strongly related to Y; assume $Y_i \approx RX_i$ (through origin).

1.1 Ratio estimators

Population ratio: $R = \tau_y/\tau_x = \mu_y/\mu_x$. Sample ratio: $\hat{R} = \bar{Y}/\bar{X}$.
Total: $\hat{\tau}_r = \hat{R}\tau_x = (\bar{Y}/\bar{X})\tau_x$.   Mean: $\hat{\mu}_r = \hat{R}\mu_x = (\bar{Y}/\bar{X})\mu_x$.
Bias exists but small for large n, so MSE $\approx$ variance.

1.2 Variance approximations

Define $Z_i = Y_i - RX_i, \sigma_r^2 = \frac{1}{N}\sum(Y_i - RX_i)^2, \hat{\sigma}_r^2 = \frac{1}{n-1}\sum(Y_i - \hat{R}X_i)^2$.
Approx design var of $\hat{\mu}_r$: $\mathrm{Var}(\hat{\mu}_r) \approx \frac{N-n}{N-1}\frac{1}{n}\sigma_r^2$.
Estimator: $\widehat{\mathrm{Var}}(\hat{\mu}_r) = \left(1 - \frac{n}{N}\right)\frac{1}{n}\hat{\sigma}_r^2$.
Total: $\mathrm{Var}(\hat{\tau}_r) \approx \frac{N-n}{N-1}\frac{N^2}{n}\sigma_r^2$, $\widehat{\mathrm{Var}}(\hat{\tau}_r) = \frac{N-n}{N}\frac{N^2}{n}\hat{\sigma}_r^2$.
Ratio R: $\mathrm{Var}(\hat{R}) \approx \frac{1}{\mu_x^2}\frac{N-n}{N-1}\frac{1}{n}\sigma_r^2$, $\widehat{\mathrm{Var}}(\hat{R}) = \frac{1}{\mu_x^2}\left(1 - \frac{n}{N}\right)\frac{1}{n}\hat{\sigma}_r^2$.

1.3 When is ratio better than SRS?

SRS mean: $\bar{Y}$, $\widehat{\mathrm{Var}}(\bar{Y}) = \left(1 - \frac{n}{N}\right)\frac{\hat{\sigma}_y^2}{n}$. Ratio mean: $\hat{\mu}_r$, $\widehat{\mathrm{Var}}(\hat{\mu}_r) = \left(1 - \frac{n}{N}\right)\frac{\hat{\sigma}_r^2}{n}$.
Key identity: $\hat{\sigma}_r^2 = \hat{\sigma}_y^2 + \hat{R}^2\hat{\sigma}_x^2 - 2\hat{R}\hat{\rho}\hat{\sigma}_x\hat{\sigma}_y$. $\hat{\rho} = \hat{\sigma}_{xy}/(\hat{\sigma}_x\hat{\sigma}_y)$
Ratio better if $\hat{\sigma}_r^2 \ll \hat{\sigma}_y^2$. Sufficient condition: $\hat{\rho} \gg \frac{1}{2}\hat{cv}_x/(\hat{cv}_y)$. or When $\hat{c}_{v,x} \approx \hat{c}_{v,y}$, "$\hat{\rho} > 0.5$ usually enough".

1.4 CI & sample size

CI: Point Estimator $\pm\, t_{n-1,1-\alpha/2}\sqrt{\widehat{\mathrm{Var}}(E)}$.
Planning (approx): $d = z_{1-\alpha/2}\sqrt{\mathrm{Var}(E)}$ with assumed σ_r^2, or iteratively solve $d = t_{n-1,1-\alpha/2}\sqrt{\widehat{\mathrm{Var}}(E)}$ for n.

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2. Ratio Estimation in Cluster Sampling

Notation (clusters)

N #clusters; M_i cluster size; $M = \sum_{j=1}^N M_j$ total elements; Y_i total Y in cluster i; $\bar{Y}_i$ sample mean in cluster i (if 2-stage); $\hat{Y}_i = M_i\bar{Y}_i$
est. cluster total; A_i count with a characteristic in cluster i; p pop proportion; $\hat{p}_i$ cluster sample proportion.

Auxiliary variable: M_i. Ratio good when Y_i and M_i are highly correlated (cluster total $\approx$ proportional to size).

2.1 One-stage clusters, total τ_y

Pop: N clusters, M known. Sample: SRS of n clusters, observe (M_i, Y_i).
Sample means: $\bar{Y} = \frac{1}{n}\sum Y_i$, $\bar{M} = \frac{1}{n}\sum M_i$.
Ratio est of total: $\hat{\tau}_r = \frac{\bar{Y}}{\bar{M}}M = \hat{R}M$, where $\hat{R} = \bar{Y}/\bar{M}$.
Variance est: $\widehat{\mathrm{Var}}(\hat{\tau}_r) = N(N-n)\frac{1}{n}\hat{\sigma}_r^2, \hat{\sigma}_r^2 = \frac{1}{n-1}\sum(Y_i - \hat{R}M_i)^2$.

2.2 One-stage clusters, proportion p

Let A_i the number of people in the cluster i who have a certain characteristic.
Ratio est of p: $\hat{p}_r = \sum_{i=1}^N A_i / \sum_{i=1}^N M_i$.
Var est: $\widehat{\mathrm{Var}}(\hat{p}_r) = \dfrac{N-n}{N}\dfrac{N^2}{M^2}\dfrac{\hat{\sigma}_r^2}{n}$, $\hat{\sigma}_r^2 = \dfrac{1}{n-1}\sum(A_i - \hat{p}_r M_i)^2$.

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3. Ratio Estimation in Two-stage Cluster Sampling

Notation (two-stage)

N clusters; cluster i has M_i elements, $M = \sum_{i=1}^N M_i$; stage 1: SRS of n clusters; stage 2 in cluster i: SRS of m_i elements; Y_{ij} value for element j in cluster i; $\bar{Y}_i$ sample mean in cluster i; $\hat{Y}_i = M_i\bar{Y}_i$; for proportions: $\hat{p}_i$ within-cluster prop. Within-cluster var: $\hat{\sigma}_i^2 = \frac{1}{m_i-1}\sum(Y_{ij} - \bar{Y}_i)^2$ (or $\hat{\sigma}_i^2 = \frac{m_i}{m_i-1}\hat{p}_i(1-\hat{p}_i)$ for binary).

3.1 Two-stage: mean μ_y

Ratio est: $\hat{\mu}_r = \dfrac{\sum_{i=1}^n M_i\bar{Y}_i}{\sum_{i=1}^n M_i} = \dfrac{\sum_{i=1}^n \hat{Y}_i}{\sum_{i=1}^n M_i}$.
Between-cluster var est: $\hat{\sigma}_r^2 = \dfrac{1}{n-1}\sum(\hat{Y}_i - \hat{\mu}_r M_i)^2$.
Design var est: $\widehat{\mathrm{Var}}(\hat{\mu}_r) = \dfrac{1}{M^2}N(N-n)\dfrac{1}{n}\hat{\sigma}_r^2 + \frac{1}{M^2}\dfrac{N}{n}\sum_{i=1}^n M_i(M_i - m_i)\dfrac{1}{m_i}\hat{\sigma}_i^2$.

3.2 Two-stage: total τ_y and proportion p

Total: $\hat{\tau}_r = M\dfrac{\sum \hat{Y}_i}{\sum M_i}$,
$\widehat{\mathrm{Var}}(\hat{\tau}_r) = N(N-n)\dfrac{1}{n}\hat{\sigma}_r^2 + \dfrac{N}{n}\sum_{i=1}^n M_i(M_i - m_i)\dfrac{1}{m_i}\hat{\sigma}_i^2$ $\hat{\sigma}_r^2 = \dfrac{1}{n-1}\sum(\hat{Y}_i - \hat{\mu}_r M_i)^2$.

Proportion: $\hat{p}_r = \dfrac{\sum M_i\hat{p}_i}{\sum M_i}$.
Var est: $\widehat{\mathrm{Var}}(\hat{p}_r) = \dfrac{N(N-n)}{nM^2}\hat{\sigma}_r^2 + \dfrac{N}{nM^2}\sum_{i=1}^n M_i(M_i - m_i)\dfrac{1}{m_i}\hat{\sigma}_i^2$,
$\hat{\sigma}_r^2 = \dfrac{1}{n-1}\sum(M_i\hat{p}_i - \hat{p}_r M_i)^2$, $\hat{\sigma}_i^2 = \dfrac{m_i}{m_i - 1}\hat{p}_i(1-\hat{p}_i)$.

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4. Regression Estimation (SRS)

Notation (regression)

known aux: μ_x, τ_x; $N, n, \hat{\rho}$ as before.
Regression = generalisation of ratio (allows non-zero intercept). Use when (X, Y) approx linear but not through origin.

4.1 LS estimators

$$\hat{b} = \frac{\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})}{\sum_{i=1}^{n}(X_i - \bar{X})^2} = \frac{\sum_{i=1}^{n} X_i Y_i - \frac{1}{n}(\sum_{i=1}^{n} X_i)(\sum_{i=1}^{n} Y_i)}{\sum_{i=1}^{n} X_i^2 - \frac{1}{n}(\sum_{i=1}^{n} X_i)^2},$$

$\hat{a} = \bar{Y} - \hat{b}\bar{X}, \hat{Y}_i = \hat{a} + \hat{b}X_i.$

4.2 Regression estimator for mean μ_y

Assume μ_x known.

Estimator: $\hat{\mu}_L = \hat{a} + \hat{b}\mu_x = \bar{Y} + \hat{b}(\mu_x - \bar{X})$. Var est: $\widehat{\mathrm{Var}}(\hat{\mu}_L) = \dfrac{N-n}{Nn}\dfrac{1}{n-2}\sum_{i=1}^{n}(Y_i - \hat{a} - \hat{b}X_i)^2$.

Approx (large n): $\mathrm{Var}(\hat{\mu}_L) \approx \dfrac{N-n}{Nn}\sigma_y^2(1-\hat{\rho}^2)$. CI: $\hat{\mu}_L \pm t_{n-2,1-\alpha/2}\sqrt{\widehat{\mathrm{Var}}(\hat{\mu}_L)}$

4.3 Regression estimator for total τ_y

Estimator: $\hat{\tau}_L = \hat{a}N + \hat{b}\tau_x$. Var est: $\widehat{\mathrm{Var}}(\hat{\tau}_L) = N(N-n)\dfrac{1}{n}\dfrac{1}{n-2}\sum(Y_i - \hat{a} - \hat{b}X_i)^2$.

4.4 SRS vs ratio vs regression (quick comparison)

- SRS: $\bar{Y}$ (or $N\bar{Y}$) unbiased, no aux info.
- Ratio: assumes line through origin; best when scatter is tight around $Y \approx RX$ and X known at pop level. Often largest gain when that assumption holds.
- Regression: line with intercept; gain roughly factor $(1-\hat{\rho}^2)$ vs SRS; more flexible than ratio when origin assumption fails.